A synopsis on

Performance Analysis of Company Teams Using Sentiment Analysis

By

Manshi Bachhan 1 (16900123057)
Ankit Kumar 2 (16900123017)
Nandan Kumar 3 (16900123060)
Beauty Mondal 4 (16900123031)
Gautam Kumar Mishra 5 (16900123043)

Department- CSE, Semester- 5th

Under the guidance of
Prof / Dr. Partha Ghosh
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Abstract:

This project aims to analyze and compare the performance of various teams within a company using sentiment analysis. Teams such as Tech Team 1 and Tech Team 2 often show variations in productivity due to differences in experience, communication, and motivation. By collecting and analyzing employee feedback, project reports, and communication data, this system will identify the key factors influencing performance. The insights derived from sentiment analysis will help management make informed decisions, such as exchanging members between teams to balance experience and skill levels, ultimately improving overall productivity and workplace satisfaction.

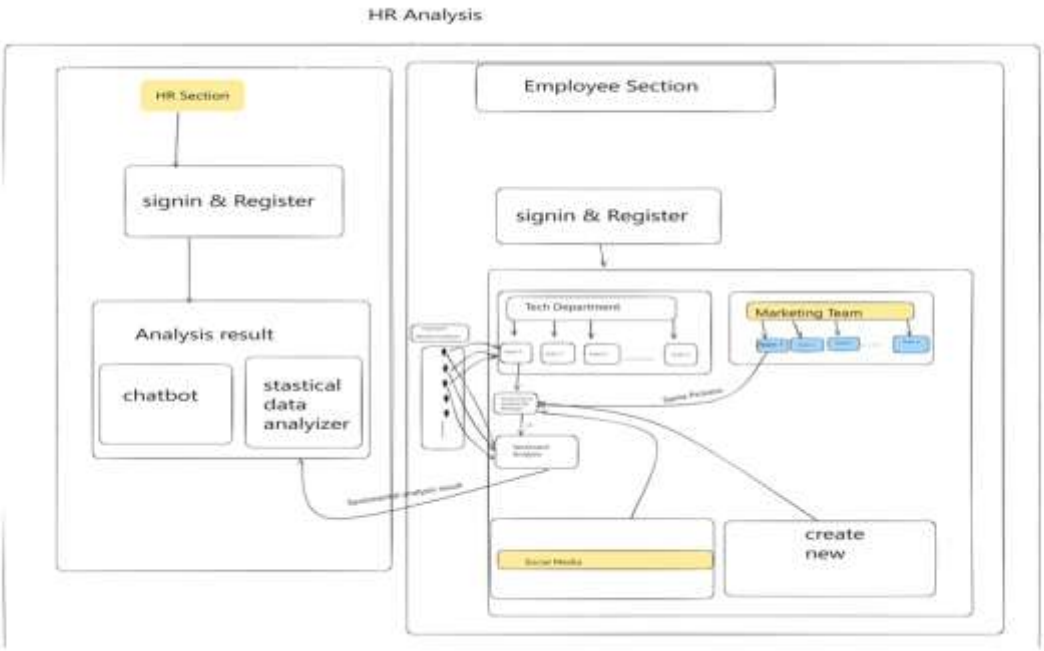
Objectives of project:

In many organizations, multiple technical teams work on similar projects but perform differently. Previous studies indicate that factors like employee satisfaction, experience, leadership, and teamwork significantly affect productivity. However, most companies lack an analytical approach to understand why one team performs better than another.

This project proposes a sentiment analysis–based system to identify reasons behind varying team performances. Employee reviews, feedback forms, and communication records will be processed using natural language processing (NLP) and machine learning models to extract sentiments (positive, negative, neutral) and detect emotional tone.

For example, if Team 1 consistently completes its projects on time with positive feedback, while Team 2 lags with negative sentiments, the model will help identify possible reasons — such as skill imbalance, lack of coordination, or low morale. Based on these findings, the system can recommend actions such as team restructuring (e.g., exchanging experienced members between teams), providing additional training, or improving management practices.

Process flow Diagram



References:

1. Dataset: kmrmanish/Employees_Reviews_Dataset · Datasets at Hugging Face

2. Model Reference: BERT Model with Fuzzy Logic Optimization on Multivariate Sentiment Analysis Tasks | Transactions on Computer Science and Intelligent Systems Research

Signature of the students

Manshi Bachhan
Ankit Kumar
Nandan Kumar
Beauty Mondal
Gautam Kumar Mishra